

RAFFLES INSTITUTION
2017 Year 6 Preliminary Examination

Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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BIOLOGY

9744/02

Paper 2 Structured Questions

19th September 2017

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/ 7
2	/ 11
3	/ 10
4	/ 12
5	/ 11
6	/ 13
7	/ 15
8	/ 10
9	/ 11
Total	/ 100

This document consists of **25** printed pages.



Raffles Institution
Internal Examination

Answer **all** the questions.

- 1 Collagen is the main structural protein in the human body. It strengthens the tendons and supports the skin and internal organs. Fig. 1.1 shows the organisation of collagen fibres.

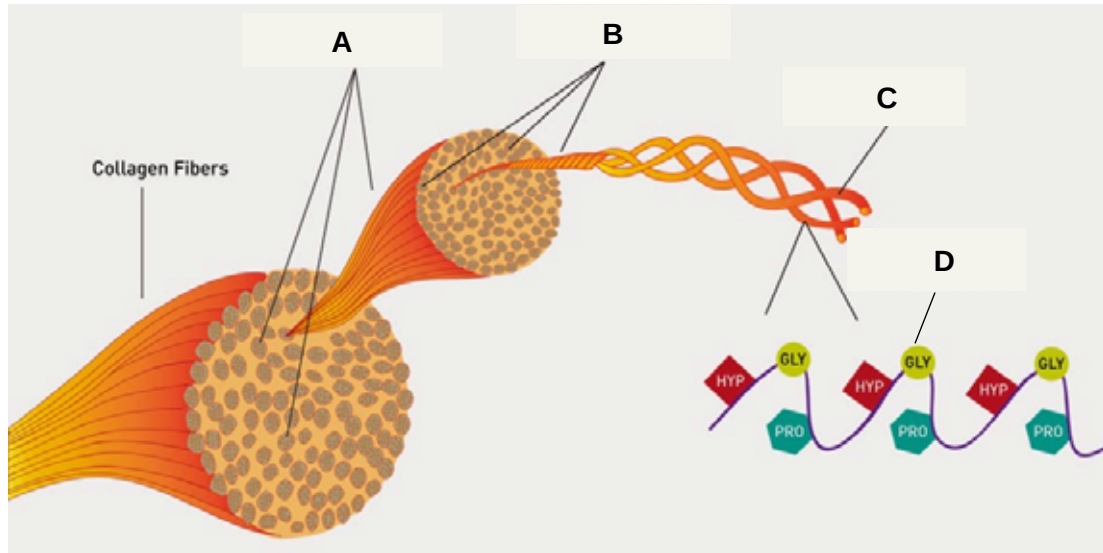


Fig. 1.1

- (a) Label structures **A**, **B**, **C** and **D** in Fig. 1.1.

A
 B
 C
 D [2]

- (b) With reference to Fig. 1.1, describe the bonds involved in the formation of collagen which contribute to its high tensile strength.

.....

 [2]

- (c) Certain pathogenic bacteria such as *Clostridium histolyticum*, have collagenases which digest collagen tissue of their hosts, causing a form of tissue death known as gangrene.

The action of collagenase can be seen in Fig.1.2, illustrating the specificity of the active site in binding to a segment of collagen and eventually cleaving it into 2 segments.

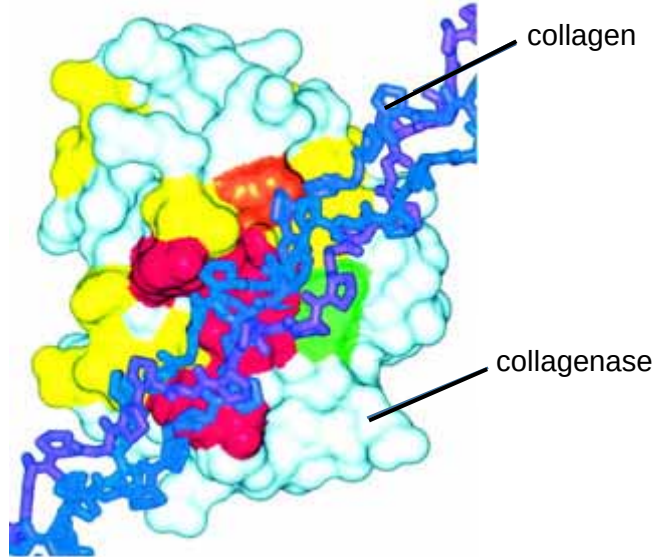


Fig. 1.2

- (i) Explain the functions of amino acids residues situated at the active site of collagenase.

.....

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.....

.....[2]

- (ii) Santyl® Ointment is an enzymatic ointment which contains collagenase. The enzyme collagenase is derived from *Clostridium histolyticum*.

Suggest a therapeutic use of collagenase.

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.....[1]

[Total : 7]

- 2 Plant cells may be grown in culture and examined using the technique of immunofluorescence in which antibodies are used to attach fluorescent dyes to specific molecules within the cells.

Fig. 2.1 is an immunofluorescent light micrograph of a plant cell at metaphase of mitosis.

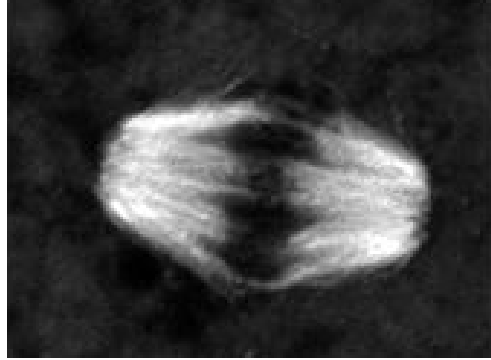


Fig. 2.1

- (a) The cell in Fig. 2.1 has two areas that are brightly stained.

- (i) Name the brightly stained structures in the two areas and outline their functions.

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.....

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.....[3]

- (ii) If a mutation occurred on the centromere of one copy of chromosome 5, suggest how this will affect mitosis.

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.....[3]

- (b) A variety of watermelon with small, sweet, seedless fruit has been produced by selective breeding. The melons also have thin skin and a uniform flavour throughout the fruit.

The selective breeding programme followed the sequence shown in Fig. 2.2.

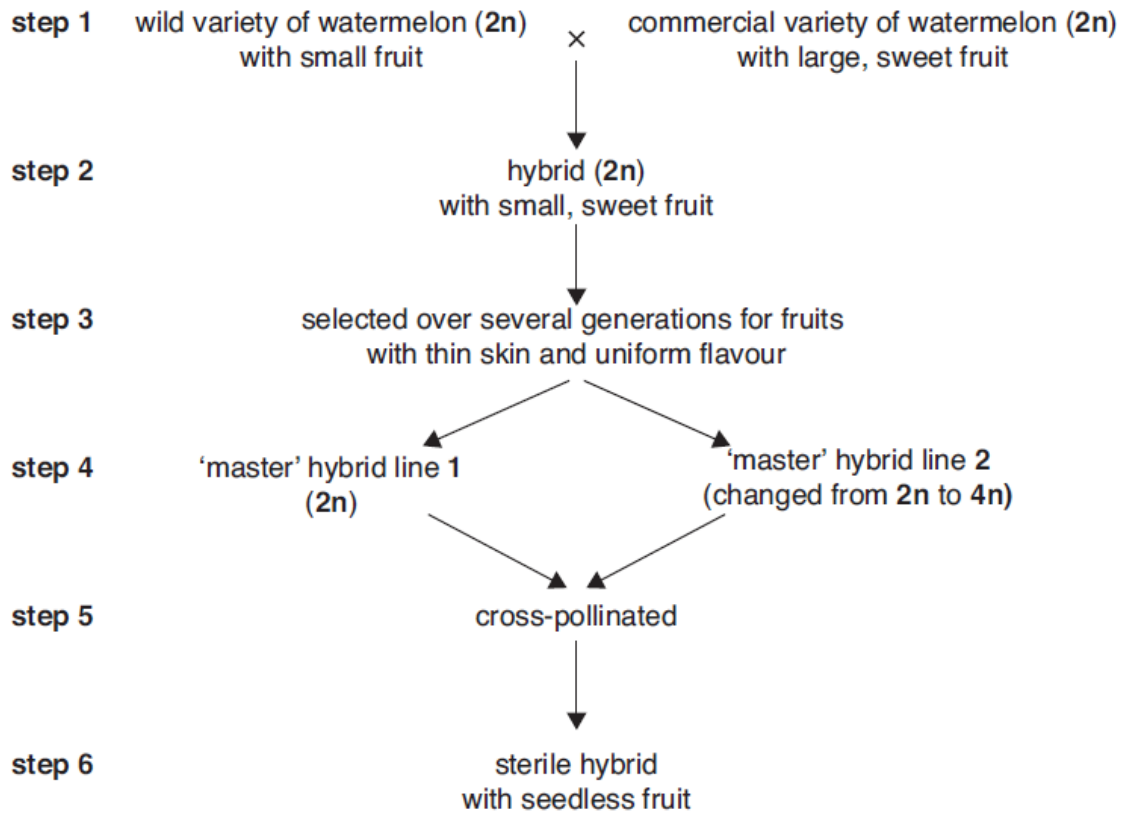


Fig. 2.2

With reference to Fig. 2.2,

- (i) explain why several generations were needed in step 3,

.....
[1]

- (ii) state what could have been used in step 4, so that the 'master' hybrid line 2 was changed from $2n$ to $4n$ and

.....[1]

(iii) explain why the hybrid produced in step 6 is sterile and seedless.

.....

.....

.....

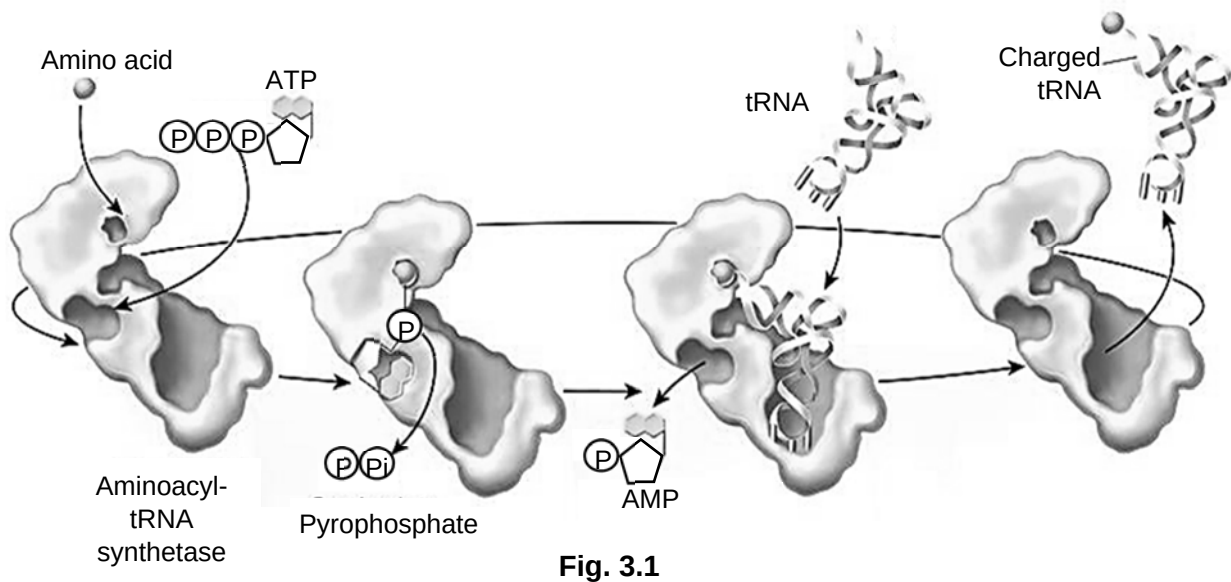
.....

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.....[3]

[Total: 11]

3 Fig. 3.1 shows the process of amino acid activation by an enzyme.



(a) Explain how the enzyme shown in the Fig. 3.1 is specific.

.....

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.....[2]

- (b) Explain why an amino acid needs to be activated.

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.....[3]

- (c) The enzyme in Fig. 3.1 exhibits the induced fit mechanism.

Explain the induced fit mechanism.

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.....[2]

The amino acid sequence of the enzyme glucose isomerase has been determined. The first five amino acids of this sequence are shown in Fig. 3.2.

Amino acid sequence :	Met – Tyr – Glu – Pro – Lys
Nucleotide sequence :	AUG UAU GAA CCU AAA

Fig. 3.2

- (d) Explain why an mRNA different from the one shown in Fig. 3.2 can result in the same amino acid sequence of Met – Tyr – Glu – Pro – Lys.

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.....[3]

[Total: 10]

- 4 In eukaryotic cells, gene expression is regulated in a highly coordinated way.

The Ras protein stimulates the cell cycle through a series of reactions. Fig. 4.1 shows a simple description of the pathway in which the Ras protein acts.

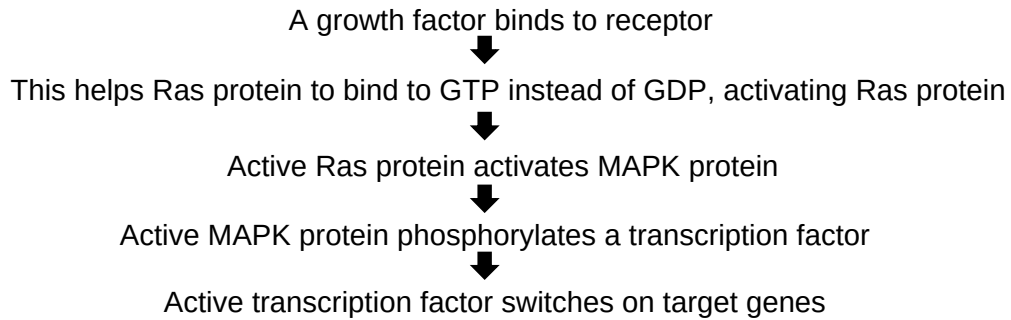


Fig. 4.1

- (a) With reference to Fig. 4.1, state the level at which the expression of these genes are regulated:

- (i) Ras gene, and

.....[1]

- (ii) the target genes.

.....[1]

- (b) Fig. 4.2 shows part of the process of RNA splicing. The snRNA (small nuclear RNA) is part of the spliceosome.

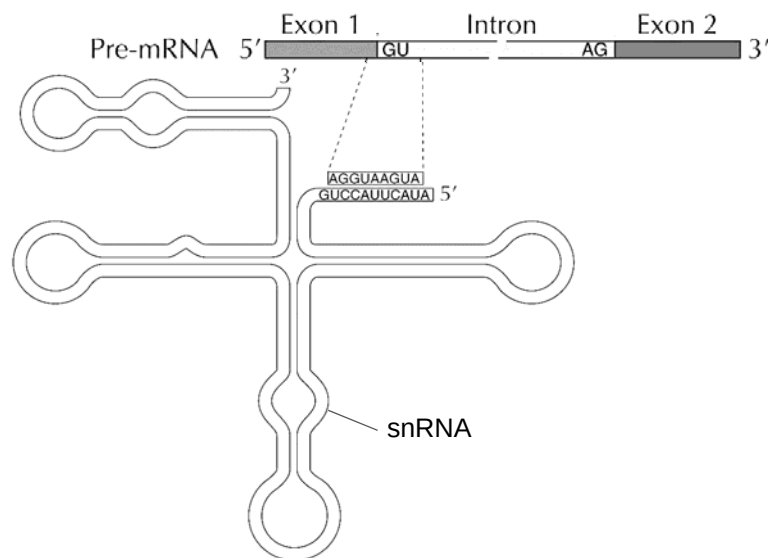


Fig. 4.2

- (i) With reference to Fig. 4.2, describe the role of snRNA.

.....

[2]

- (ii) Describe how regulation at the post-transcriptional level can be carried out via alternative splicing.

.....

 [2]

- (iii) In the study of particular tissues, different mRNAs were produced from a single gene. However only a particular mRNA was detected at high levels.

Suggest how the structure of the mRNA accounts for this.

.....
 [1]

- (c) Regulation of gene expression can also be carried out at the post-translational level.

Fig. 4.3 below shows the control gene expression using ubiquitin and proteasome.

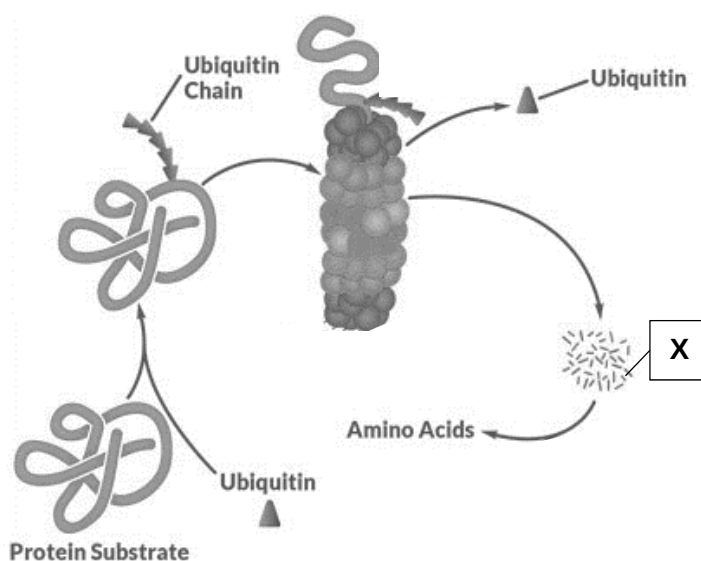


Fig. 4.3

- (i) Name molecule X.

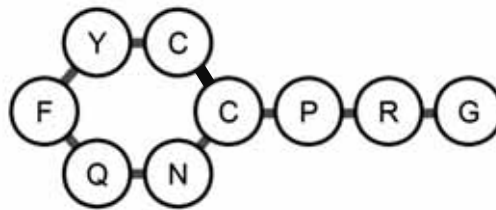
..... [1]

- (ii) With reference to Fig. 4.3, explain how cellular proteins are degraded using this system.

.....

 [3]

Fig. 4.4. represents vasopressin, a small peptide hormone, with the following structure.



Key: C = cysteine, F = phenylalanine, G = glutamic acid, P = proline, Q = glutamine, R = arginine, Y = tyrosine, N = asparagine

Fig. 4.4

- (iii) Describe one post-translational modification that must have occurred to form vasopressin.

.....[1]

[Total: 12]

- 5 Cystic fibrosis is a human disorder related to abnormally thick and sticky mucus which can lead to damage in various body organs. Fig. 5.1 shows the inheritance of cystic fibrosis in a Norwegian family. Those individuals with the condition are shaded.

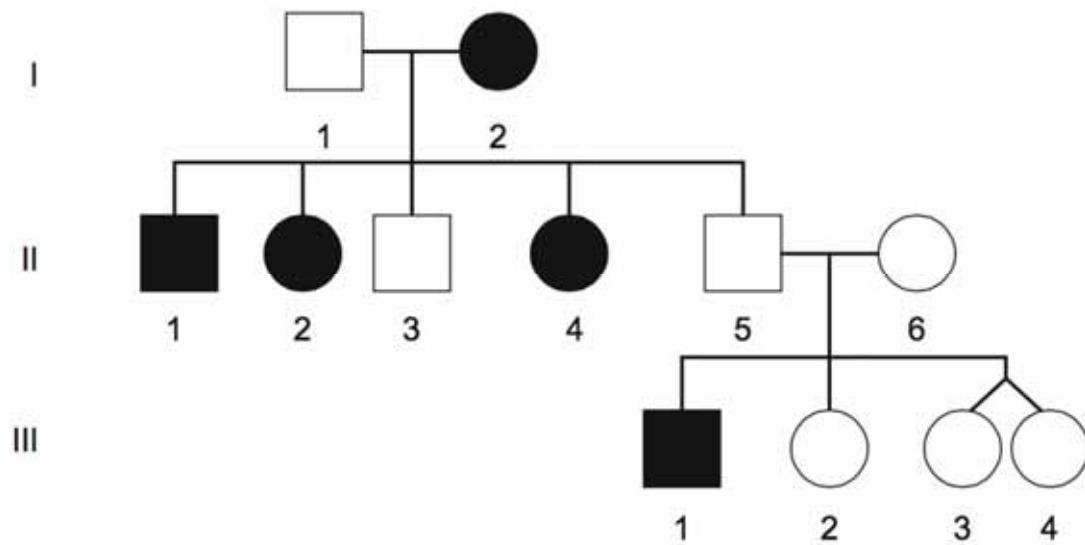


Fig. 5.1

- (a) (i) State the mode of inheritance of cystic fibrosis and give two reasons to support your answer.

Mode of inheritance

.....[1]

Reason 1

.....

.....

.....[1]

Reason 2

.....

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.....[1]

- (ii) Individuals III-3 and III-4 are identical twins. They were recruited for a study, looking for trait differences expressed by the two individuals. Of the 10 traits studied, they showed a difference in two of the traits.

Explain what the findings of such a study revealed.

.....

.....[1]

- (b) In the primula plant, the pigment malvidin creates blue-coloured flowers. Synthesis of malvidin is controlled by gene **A**. However, the production of this pigment is under the influence of another gene, gene **D**, which is found on a different chromosome. In the presence of gene product **D**, the primula plant will produce white flowers.

- (i) White flowers were observed in plants that carried both the dominant alleles A and D.

Explain the interaction between gene A and gene D in influencing the colour of the primula flower.

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.....[3]

A cross between two primula plants with white flowers, heterozygous at both gene loci, yielded the following offspring:

Primula plants bearing blue flowers: 37

Primula plants bearing white flowers: 155

(ii) Draw a genetic diagram in the space provided, to explain the above cross.

[4]

[Total: 11]

- 6 Pyruvate, a product of glycolysis, needs to move from the cytosol into the mitochondrion. After some processing, pyruvate will be converted to acetyl Co A which then enters the Krebs cycle.

Fig. 6.1 is an electron micrograph of a mitochondrion.

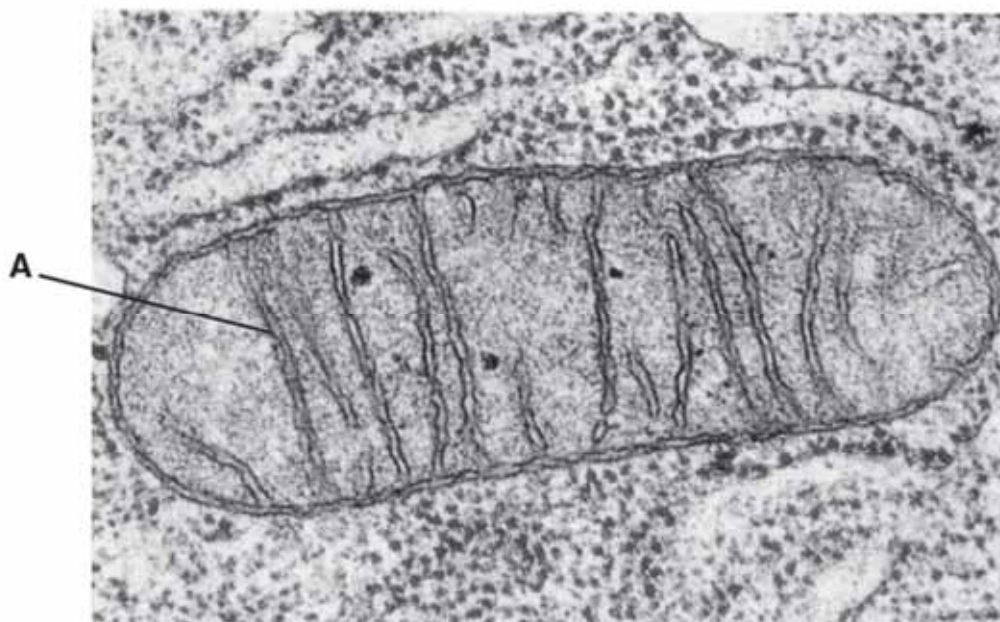


Fig. 6.1

- (a) Name the region within the mitochondrion where the Krebs cycle occurs and using the symbol X, indicate this region on Fig. 6.1.

region X:[1]

- (b) Pyruvate requires the help of pyruvate translocase to enter the mitochondrion. The structure of pyruvate is shown in Fig. 6.2.

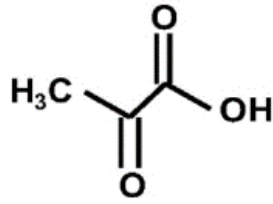


Fig. 6.2

Pyruvate translocase is located on the membrane labelled 'A' in Fig. 6.1.

Fig. 6.3 shows how pyruvate is transported across the membrane A.

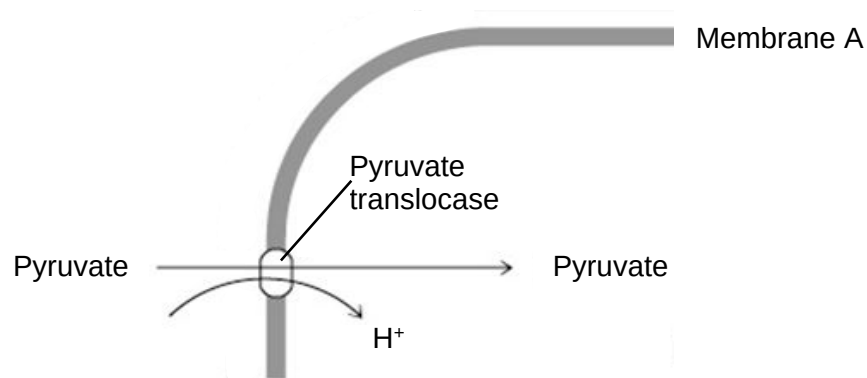


Fig. 6.3

- (i) Explain why pyruvate requires the help of pyruvate translocase to cross membrane A.

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 [2]

- (ii) With reference to Fig. 6.3, describe the transport of pyruvate across membrane A.

.....
 [1]

(c) Other than pyruvate translocase, the electron transport chain and ATP synthase can also be found on membrane A.

(i) Explain how pyruvate translocase, the electron transport chain and ATP synthase are held in membrane A.

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 [2]

(ii) Explain how mitochondria are adapted to contain many structures such as pyruvate translocase, the electron transport chain and ATP synthase.

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 [1]

(d) Fig. 6.4 shows a section of membrane A.

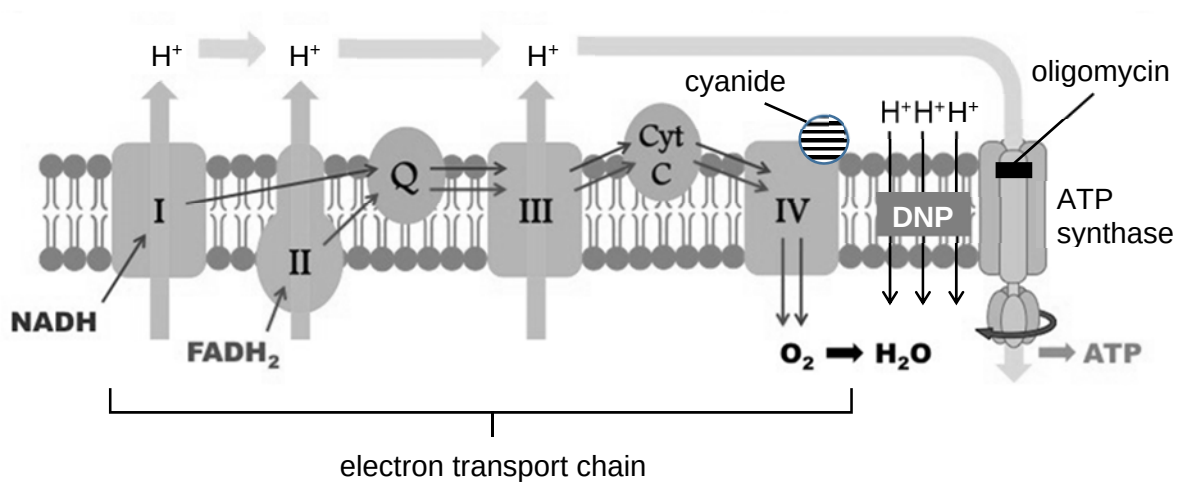


Fig. 6.4

Dinitrophenol (DNP), cyanide and oligomycin are chemicals that interfere with the normal functioning of the components shown in membrane A.

(i) DNP can shuttle H^+ across biological membranes.

With reference to Fig. 6.4, explain why fewer ATP molecules were produced when DNP was added to mitochondria.

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 [1]

- (ii) Cyanide inhibits complex IV shown in Fig. 6.4.

Explain why in the presence of cyanide, oxygen consumption decreases and lactate production increases.

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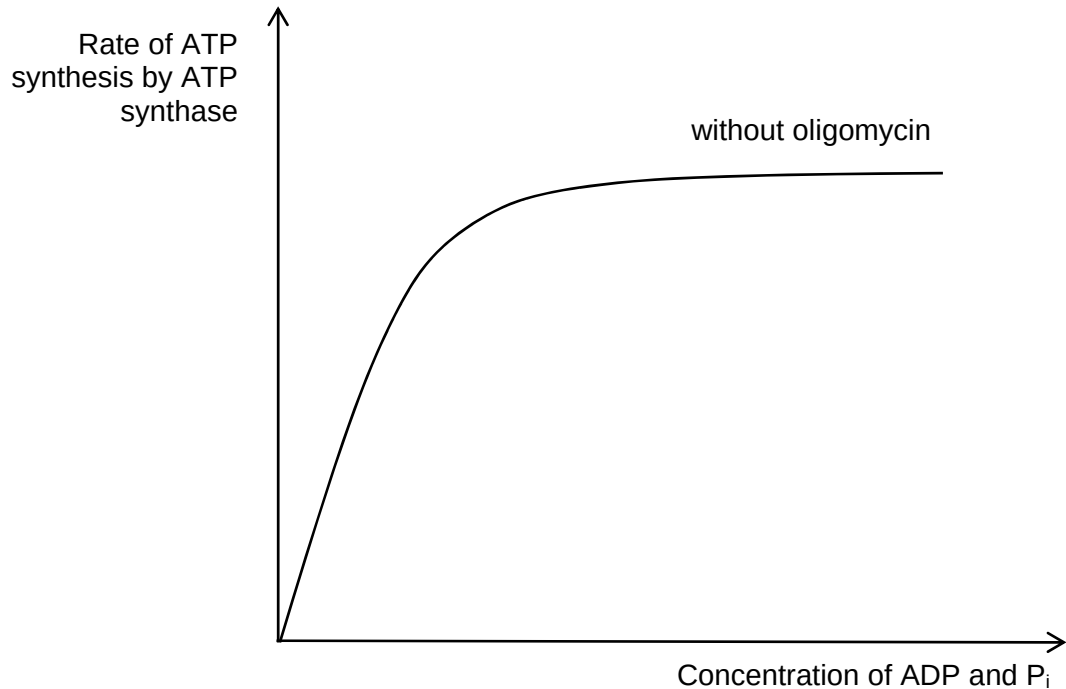
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..... [4]

- (iii) Oligomycin inhibits ATP synthase by blocking the proton channel, as seen in Fig. 6.4.

Indicate in the graph below, the effect of administering a fixed amount of oligomycin on the rate of ATP synthesis. [1]



[Total: 13]

- 7 *Streptococcus pyogenes* bacteria causes a range of diseases including skin infections and respiratory illnesses. Treatment of the diseases is carried out using antibiotics such as penicillin and erythromycin.

In 1988, a nation-wide movement to reduce the use of the antibiotic erythromycin to treat patients infected with *S. pyogenes* was started in Finland.

Fig. 7.1 shows the number of doses of erythromycin used per thousand people per month, over a period of eleven years from 1984 to 1994. The figure also shows the percentage of infections each year caused by erythromycin-resistant strains of *S. pyogenes*.

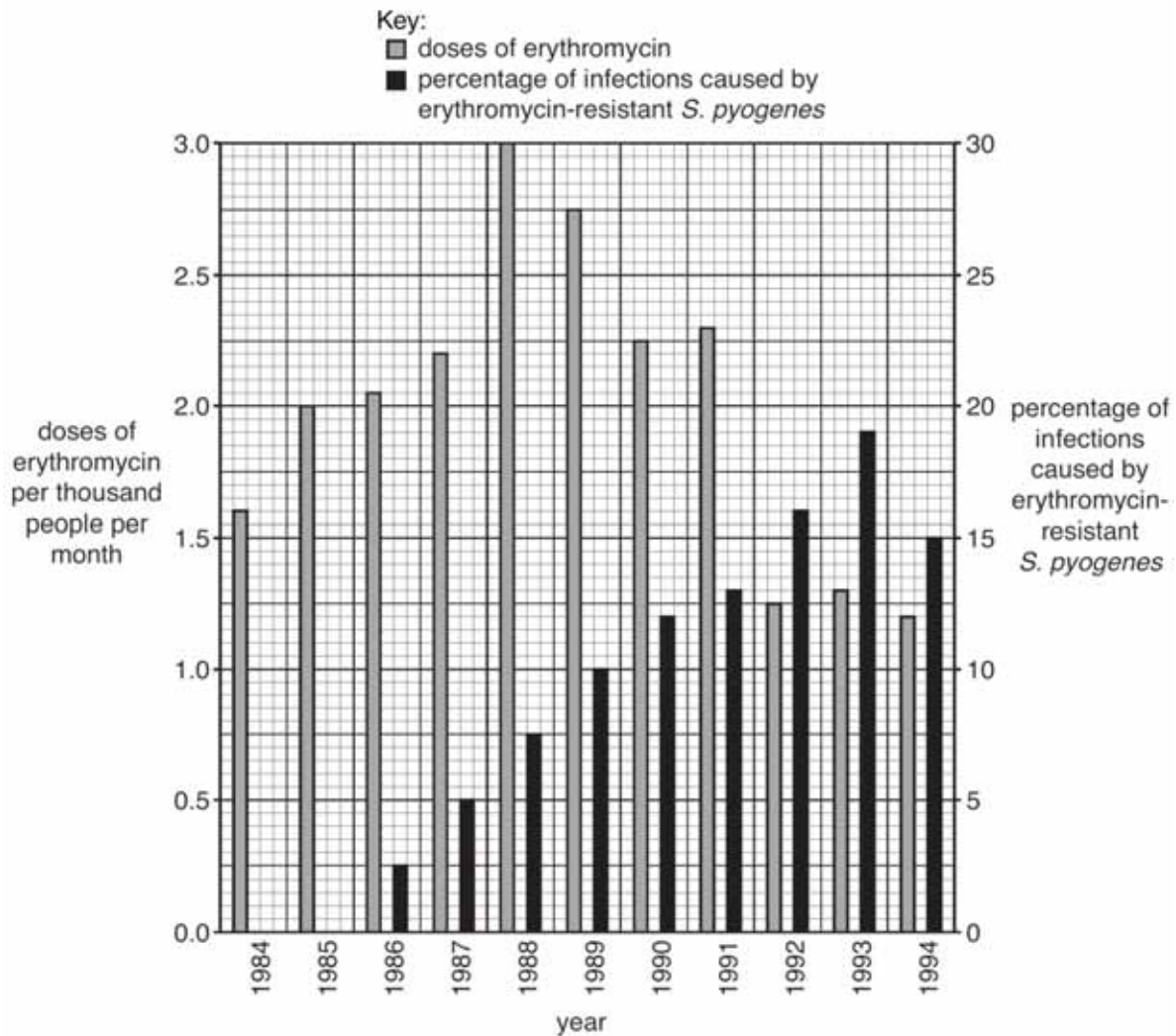


Fig. 7.1

- (a) (i) Describe two structural features that are typical of bacteria, including *S. pyogenes*.

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.....[2]

- (ii) Explain the advantages to scientists of giving the bacterium *Streptococcus pyogenes* a binomial Latin name.

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.....[2]

- (b) (i) With reference to Fig. 7.1, describe the trend in the use of erythromycin between 1984 and 1994.

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.....[3]

- (ii) Apart from mutation, suggest how the erythromycin-resistant *S. pyogenes* may have originated in Finland in 1986.

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.....[1]

- (iii) Explain why the percentage of erythromycin-resistant bacterial infections increased between 1986 and 1993.

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.....[4]

- (iv) Suggest why the percentage of erythromycin-resistant bacterial infections fell between 1993 and 1994.

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.....[2]

- (c) In 2014, the World Health Organisation highlighted the prevalence of antibiotic resistance in bacteria as a global health threat. This prompted the urgent development of alternative methods to the use of antibiotics to treat bacterial infections.

Phage therapy, which is the use of bacterial viruses to defend against pathogenic bacteria is a strategy to address this issue.

Suggest an advantage of using phage therapy.

.....

.....[1]

[Total: 15]

- 8 Sea slugs are marine molluscs and two are shown in Figs. 8.1 and 8.2. Sea slugs live in shallow coastal waters where they graze on animals that form colonies on rocks, like sponges and corals. Sea slugs are preyed upon by fish.



Fig. 8.1



Fig. 8.2

Many sea slugs collect poisonous chemicals or defenses such as stinging cells from their prey. They then make use of these in their own bodies. Sea slugs containing poisonous chemicals or other defenses are generally brightly coloured.

Sea slugs evolved from sea snails by the loss of the shell. It is thought that being brightly coloured is an advantage to poisonous animals.

- (a) Suggest why being poisonous and brightly-coloured is more common in sea slugs than in sea snails.

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.....[2]

- (b) The waters of South-Eastern Australia are home to a group of sea slugs found nowhere else in the world.

Three species of the genus *Chromodoris* all share a pattern of bright red spots on a white background. This is an example of mimicry. One of the *Chromodoris* species is shown in Fig. 8.1.

- (i) Explain why one type of warning pattern shared by three species gives a greater selective advantage to the slugs than if each had its own distinctive warning pattern.

.....

.....

.....

.....[2]

- (ii) Describe how a South-Eastern Australian sea slug population that was originally white might have developed the red-spotted pattern over a period of time.

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.....[4]

- (iii) Explain how environmental factors can act as stabilising forces of natural selection in the waters of South-Eastern Australia, after the initial evolution of the three species of sea slugs of genus *Chromodoris*.

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.....
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.....[2]

[Total: 10]

- 9 The Calvin cycle is a stage of photosynthesis during which carbon dioxide is fixed.

- (a) (i) State the products of the light dependent stage that are used in the Calvin cycle.

.....[1]

- (ii) Describe the fate of the triose sugar produced in the Calvin cycle.

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.....
.....
.....[2]

- (b) The conditions in which young plants of wheat and maize are grown affects their ability to photosynthesise at high and low temperatures when they mature.

Young maize plants were grown to maturity at high and low temperatures. When they were mature, their rate of photosynthesis was measured at different temperatures. The results are shown in Fig. 9.1.

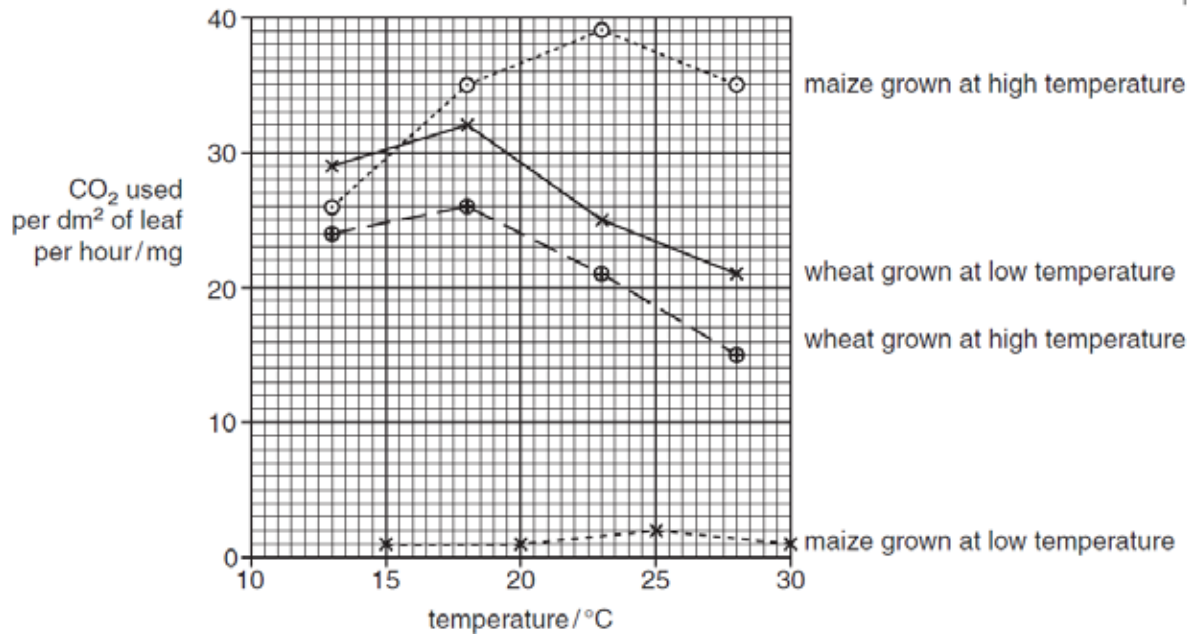


Fig. 9.1

- (i) With reference to Fig. 9.1, compare the effect of temperature on the rate of photosynthesis of the maize plants and wheat plants that were grown at high temperature when they were young.

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[2]

- (ii) Low temperatures slow down the formation of membranes inside the chloroplasts in maize leaves but not in wheat leaves.

Use this information to explain the differences between the results for maize and wheat grown at low temperatures in Fig. 9.1.

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[2]

- (iii) In view of this information, suggest how maize farmers in temperate countries should adapt their practices to ensure higher yield in the field.

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[1]

- (c) Table 9.1 shows the yield in plot tests of the same variety of wheat grown at 18°C and 20°C.

No.	Wheat grown at 18°C	Wheat grown at 20°C
1	22.9	13.7
2	19.8	18.2
3	24.4	17.5
4	27.9	15.1
5	23.1	21.6
6	25.7	19.2
7	28.2	21.6
8	25.6	24.8
9	28.7	25.2
10	31.5	27.8
11	26.2	25.2
12	37.0	34.0
	Mean = 26.75	Mean = 21.99

Table 9.1

- (i) Given that the t-test value is 2.26, examine the data and use the information given on the next page to decide whether or not the wheat grown at the 2 temperatures are different in yielding ability.

.....

[2]

- (ii) In view of global warming, suggest an implication of the finding in c(i).

.....
[1]

Degrees of freedom	Significance level					
	20% (0.20)	10% (0.10)	5% (0.05)	2% (0.02)	1% (0.01)	0.1% (0.001)
1	3.078	6.314	12.706	31.821	63.657	636.619
2	1.886	2.920	4.303	6.965	9.925	31.598
3	1.638	2.353	3.182	4.541	5.841	12.941
4	1.533	2.132	2.776	3.747	4.604	8.610
5	1.476	2.015	2.571	3.365	4.032	6.859
6	1.440	1.943	2.447	3.143	3.707	5.959
7	1.415	1.895	2.365	2.998	3.499	5.405
8	1.397	1.860	2.306	2.896	3.355	5.041
9	1.383	1.833	2.262	2.821	3.250	4.781
10	1.372	1.812	2.228	2.764	3.169	4.587
11	1.363	1.796	2.201	2.718	3.106	4.437
12	1.356	1.782	2.179	2.681	3.055	4.318
13	1.350	1.771	2.160	2.650	3.012	4.221
14	1.345	1.761	2.145	2.624	2.977	4.140
15	1.341	1.753	2.131	2.602	2.947	4.073
16	1.337	1.746	2.120	2.583	2.921	4.015
17	1.333	1.740	2.110	2.567	2.898	3.965
18	1.330	1.734	2.101	2.552	2.878	3.922
19	1.328	1.729	2.093	2.539	2.861	3.883
20	1.325	1.725	2.086	2.528	2.845	3.850
21	1.323	1.721	2.080	2.518	2.831	3.819
22	1.321	1.717	2.074	2.508	2.819	3.792
23	1.319	1.714	2.069	2.500	2.807	3.767
24	1.318	1.711	2.064	2.492	2.797	3.745
25	1.316	1.708	2.060	2.485	2.787	3.725
26	1.315	1.706	2.056	2.479	2.779	3.707
27	1.314	1.703	2.052	2.473	2.771	3.690
28	1.313	1.701	2.048	2.467	2.763	3.674
29	1.311	1.699	2.043	2.462	2.756	3.659
30	1.310	1.697	2.042	2.457	2.750	3.646
40	1.303	1.684	2.021	2.423	2.704	3.551
60	1.296	1.671	2.000	2.390	2.660	3.460
120	1.289	1.658	1.980	2.158	2.617	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.291

$$v = n_1 + n_2 - 2$$

where v = degrees of freedom

n = sample size (number of observations)

[Total:11]

End of Paper